Bonus 5: The Double Pendulum

Week 5 — two coupled equations, one Euler loop, and a figure

Submission: paste `bonus05.jl` into the Week 5 bonus CodeRunner question on Moodle, and upload `double_pendulum.png` with the bonus figure submission.

Optional bonus

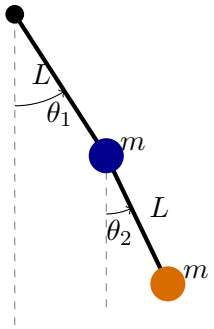
This bonus is optional and is not a prerequisite for later work. Nothing that appears *only* here is examined; ideas that also belong to the core lectures can still be assessed. It is worth **one-third of a final-grade point**. It uses only functions, loops, vectors, and plotting from this week and earlier.

Read first: the two-link pendulum

Hang one pendulum from a fixed pivot, and hang a second pendulum from the first bob. Give the upper rod a gentle push while the lower one starts at rest. The lower bob does not simply follow: it starts to swing, and its motion feeds back on the upper one. This is the double pendulum.

We take both rods to be rigid and massless, both bobs to have mass m, and both rods to have length L. Angles are measured from the downward vertical, positive counterclockwise. The two bobs sit at

$$x_1 = L\sin\theta_1, \quad y_1 = -L\cos\theta_1, \qquad x_2 = x_1 + L\sin\theta_2, \quad y_2 = y_1 - L\cos\theta_2.$$



With $\omega_1 = \dot\theta_1$ and $\omega_2 = \dot\theta_2$, the kinetic and potential energies are

$$T = mL^2\left(\omega_1^2 + \tfrac{1}{2}\omega_2^2 + \omega_1\omega_2\cos(\theta_1 - \theta_2)\right), \qquad V = -mgL\left(2\cos\theta_1 + \cos\theta_2\right).$$

Applying the Euler–Lagrange equations—the formalism you will meet next year in Analytical Mechanics—to these two energies gives one vector equation for the four state variables:

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix}\theta_1\\ \theta_2\\ \omega_1\\ \omega_2\end{pmatrix} = \begin{pmatrix} \omega_1 \\ \omega_2 \\ \dfrac{-3g\sin\theta_1 - g\sin(\theta_1 - 2\theta_2) - 2\sin(\theta_1-\theta_2)\left[L\omega_2^2 + L\omega_1^2\cos(\theta_1-\theta_2)\right]}{L\left[3 - \cos(2\theta_1 - 2\theta_2)\right]} \\ \dfrac{2\sin(\theta_1-\theta_2)\left[2L\omega_1^2 + 2g\cos\theta_1 + L\omega_2^2\cos(\theta_1-\theta_2)\right]}{L\left[3 - \cos(2\theta_1 - 2\theta_2)\right]} \end{pmatrix}.$$

The top two components are the angle rates; the bottom two are the angular accelerations. Every term is a number once you know $\theta_1, \theta_2, \omega_1, \omega_2, L, g$, and the denominator is common

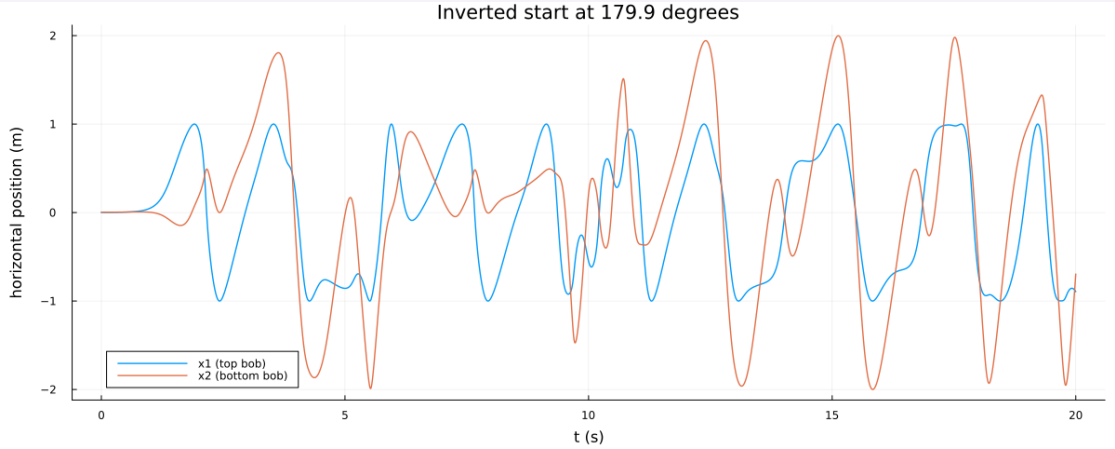
to both accelerations. There is no small-angle approximation: the sines use the actual angles, so the motion is genuinely nonlinear once the amplitudes grow.

Physics box: what the model assumes

Rigid massless rods, point masses, a frictionless pivot, uniform gravity $g > 0$ in $\mathrm{m/s^2}$, and no air resistance. Angles are in radians and angular velocities in rad/s. Both masses and both lengths are equal, which is why the equations above are shorter than the general two-mass two-length version. The exact continuous motion conserves total mechanical energy $E = T + V$. Euler integration does not: it slowly drifts, which is acceptable for this programming exercise but not for a careful long-time study.

Small displacements stay in the linear regime. There the two rods behave like two coupled oscillators with two normal-mode frequencies, and their sum produces the slow beat visible in the figure: energy passes back and forth between the modes while the motion remains predictable. This is the regime used in the task and in the supplied checks.

Large displacements behave differently. Start both rods pointing up, $\theta_1 = \theta_2 = 180°$, and the configuration is an unstable equilibrium: it balances when exactly at rest, and in floating point it eventually topples. Nudge it instead—release at $179.9°$, or give the upper rod a small angular velocity there—and the fall that follows is chaotic. Two runs that differ by 10^{-9} rad end up at angles differing by many radians within about 20 s, whereas the same 10^{-9} difference at the small-amplitude start is still 10^{-9}. That sensitivity is real physics, not a bug in the code, but it means a chaotic trajectory cannot be compared against a reference. The equations you implement are checked exactly; only the small-amplitude figure is inspected.



The panel above shows the horizontal positions x_1 and x_2 of the two bobs for the $179.9°$ release. Nothing repeats, and no beat survives: this is the chaotic regime.

Task

Write a function for the right-hand side,

```
function double_pendulum_rhs(state; length = 1.0, gravity = 9.81)
    # state is [theta1, theta2, omega1, omega2]
    # return [dtheta1/dt, dtheta2/dt, domega1/dt, domega2/dt]
end
```

and an Euler integration, in the same style as the projectile homework:

$$\text{state}_{\text{next}} = \text{state}_{\text{now}} + \Delta t \cdot \texttt{double_pendulum_rhs}(\text{state}_{\text{now}}).$$

Store the history in a matrix with columns

$$\texttt{[t, theta1, theta2, omega1, omega2]}.$$

Start the system from rest in the following sense: both rods hang straight down, $\theta_1(0) = \theta_2(0) = 0$, the lower bob has no angular velocity, $\omega_2(0) = 0$, and the upper rod receives a *small* initial angular velocity

$$\omega_1(0) = 0.05 \text{ rad/s}.$$

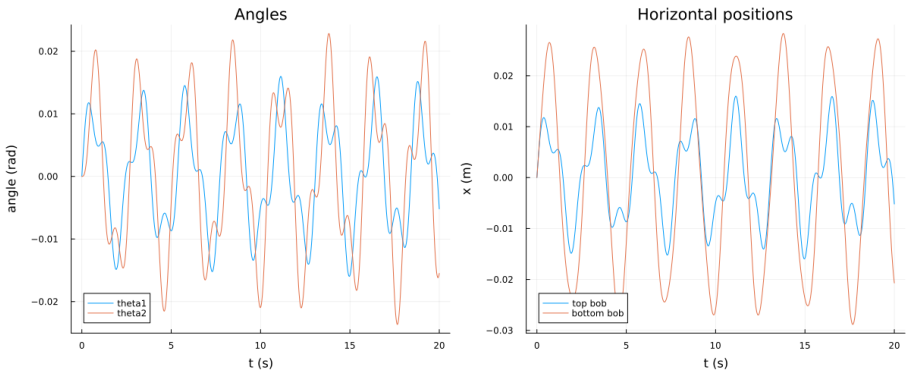
Integrate to $T = 20$ s with `dt = 0.001` s. Assume sensible inputs; no error checking is needed.

A figure to hand in

Save one figure, `double_pendulum.png`, with two panels. The left panel shows $\theta_1(t)$ and $\theta_2(t)$ on the same axes; the right panel shows the horizontal positions $x_1(t)$ and $x_2(t)$ of the two bobs, so that the small displacements are easy to see. The following supplied code assumes your history matrix is called `history` and that you have already run `using Plots`:

```
let
    rod_length = 1.0
    times = history[:, 1]
    theta1 = history[:, 2]
    theta2 = history[:, 3]
    x1 = rod_length .* sin.(theta1);  y1 = -rod_length .* cos.(theta1)
    x2 = x1 .+ rod_length .* sin.(theta2);  y2 = y1 .- rod_length .* cos.(theta2)
    p1 = plot(times, theta1; label = "theta1", xlabel = "t (s)",
        ylabel = "angle (rad)", title = "Angles")
    plot!(p1, times, theta2; label = "theta2")
    p2 = plot(times, x1; label = "top bob", xlabel = "t (s)",
        ylabel = "x (m)", title = "Horizontal positions")
    plot!(p2, times, x2; label = "bottom bob")
    p = plot(p1, p2; layout = (1, 2), size = (1200, 480),
        left_margin = 5 * Plots.mm, bottom_margin = 4 * Plots.mm)
    savefig(p, "double_pendulum.png")
end
```

The lower bob should visibly swing although it starts with no velocity of its own. Expect angles of a few hundredths of a radian.



The example uses the stated initial condition with $L = 1$ m and $g = 9.81$ m/s^2. Your curves need not match it pixel for pixel, but the scale and qualitative behavior should agree.

Values to check the equations

The equations are easy to mistype, so check the right-hand side directly at a few states before integrating. With $L = 1$ m and $g = 9.81$ m/s^2:

```
println(double_pendulum_rhs([0.1, 0.0, 0.0, 0.0]))
println(double_pendulum_rhs([0.0, 0.0, 0.05, 0.0]))
println(double_pendulum_rhs([0.2, -0.1, 0.3, -0.2]))
```

```
[0.0, 0.0, -1.9394021734809634, 1.929713240762815]
[0.05, 0.0, -0.0, 0.0]
[0.3, -0.2, -4.47953801481895, 5.285428755886728]
```

The first case has both rods at rest with the upper rod displaced; the upper bob is pulled back while the lower one is pushed forward. The second has both rods vertical with only the upper one moving, so both accelerations vanish. The third exercises every term at once.

Submission and scope

Submit `bonus05.jl` with `double_pendulum_rhs` and the integration program, plus `double_pendulum.png`. The check calls your function at supplied states within a small tolerance; the figure is inspected for the required panels and behavior. No report is required.

This bonus is independent of HW 5 and is not a prerequisite. Nothing that appears only here is examined.